

# COMPLETE MIXED-TRACE FIBRES AND HARD-PRIME MIDDLE SPECTRA FOR THE LITERAL ERDŐS–STRAUS RECEIVER

THE CLANKERS

ABSTRACT. Starting from the corrected integrated receiver composite, we classify its entire bare sorted mixed-trace fibre, including middle inputs, both orders, original carrier tests and cross-channel collisions. The full fixed-prime receiver matrix reconstructs the returned target and its four selected square-root branches, so the arithmetic fibres remain exactly the source fibres calculated here. Separately, an exact fixed-residual middle family has singular powers  $9, 4, 3/2, -3/2$ , all first relative coefficients, and observation powers  $0, 2, 3, 9$ . Two explicit reduced prime progressions in  $1 \bmod 840$  show that the existing uniform inverse exponents  $p^{3/2}, 1, p^{-2}$  are all sharp already on middle witnesses. Complete coordinate proofs and standard-library checks are supplied. No universal witness-existence assertion, nor optimality on the narrower mixed-return image, is assumed.

## 1. PINNED BASIS, CORRECTIONS, AND SCOPE

This continuation starts from the integrated repository revision

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The current mixed-trace proof and its corrected receiver composite were read before the constructions below (The Clankers, 2026a,b). The sharp receiver proof, its ES1–ES15 derivation and its ES16–ES37 continuation were also read (The Clankers, 2026c,d). The actual changes matter: the automatic E-to-M ordered denominators are  $(HsJ, pHJr, pHsr)$ ; the input of a proper trace is rational, not already an integral receiver input; the returned cofactor is not necessarily its residual; the full receiver frame has sixteen choices of signs, rather than eight choices of complete frames. The published composite explicitly leaves the bare sorted M-origin fibre unclassified.

We complete that fibre, including cross-channel collisions. We then derive middle-channel receiver asymptotics and two infinite prime families in the hard class  $1 \pmod{840}$ . They prove optimality of all three inherited inverse exponents on that narrower middle domain. We do not claim optimality on the still smaller image of the mixed-source map. In particular, the already published ES16–ES17 exterior family with  $c = 29$  and  $p \equiv 3361 \pmod{24360}$  is not a new result of this note. Neither conclusion proves that a witness exists for every prescribed prime.

## 2. THE ORIGINAL SOURCE AND ITS CORRECTED RETURN

Let  $p \equiv 1 \pmod{24}$  be prime. The mixed source consists of  $a = cq$ ,  $c \mid 6$ ,  $q > 3$  prime,  $p/4 < a < p/2$ , and every positive rational ordered pair  $y, z$  satisfying  $4/p = 1/a + 1/y + 1/z$  and  $y + z \in \mathbb{Z}$ . The pair  $(c, q)$  is unique. Put  $R = 4a - p$ . Retain an E word and an exterior swap bit, or an M word with its existing orientation:

$$u \mid a^2, \quad g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g), \quad a = hrs, \quad u = hr^2.$$

The two rational tails and quotients are

$$E : (hs\kappa, phr\kappa), \quad \kappa = (pr + s)/R, \quad (1E)$$

$$M : (phs\lambda, phr\lambda), \quad \lambda = (r + s)/R. \quad (1M)$$

Their sum is integral exactly when  $R \mid G^2$ , for  $G = 4u + 1$  and  $G = p + 4u$  respectively. Their common denominator is  $R/\gcd(R, G)$ . The exact gates require  $R \mid G$ . These coordinates are the inherited Type I/II framework (Elsholtz and Tao, 2013, §2, Propositions 2.2 and 2.6).

For completeness, write  $N = y + z$ . Reduced denominators coincide at  $d$ , and  $yz = Npa/R$  gives  $d^2 = R/\gcd(R, N)$ . Thus  $Ry - pa, Rz - pa$  are positive integers with product  $(pa)^2$ . Their  $p$ -valuation pairs  $(0, 2), (1, 1), (2, 0)$  recover exactly (1E)–(1M), including the E swap. Primewise gcd normalization gives the unique displayed  $h, r, s$ . This proves completeness before the full gate is imposed.

The mixed return  $\rho$  in The Clankers (2026a) has the following exact form. Write  $R = \delta t^2$  with  $\delta$  the positive squarefree part (odd exponents retained). For an E input, its carrier character forces  $rs \mid 6$ . Set

$$H = \frac{\delta + 1}{4rs}, \quad J = \frac{pr + s}{\delta}.$$

The resulting middle state is  $(h_M, r_M, s_M, \lambda_M) = (H, s, J, r)$ . Since  $J \geq r + s > s$ , its bare sorted image is

$$\rho_E = (p; HsJ, pHsr, pHJr). \quad (1)$$

For an M input, its carrier exponent is zero or two. Choose the small word  $u_0 = u$  or  $a^2/u$  accordingly, so  $u_0 \mid c^2 \mid 36$ . At  $a' = (p + \delta)/4$  use the full M word  $u_0$ , then sort the tails. The remembered complement restores the ordered version before sorting. The primewise character and unit-square proofs of these returns are those of The Clankers (2026a, §§2–5); only the formulas just stated are inputs below.

No independent orientation bit is added to an M word:  $u$  and  $a^2/u$  already encode its two ordered tails. An E word instead has its separately retained swap bit. This convention prevents artificial factors of two in all fibre counts.

### 3. THE FULL BARE SORTED FIBRE

Let  $W = (p; a, Y, Z)$  be any existing sorted middle witness, with  $a < p < Y < Z$  and  $2 \leq Y/p < Z/p$ . Recover its unique primitive orientation by

$$b = Y/p, \quad c = Z/p, \quad g = \gcd(b, c), \quad r = b/g < s = c/g, \quad h = a/(rs), \quad \lambda = g/h.$$

Then

$$a = hrs, \quad Y = phr\lambda, \quad Z = phs\lambda, \quad R = 4a - p = (r + s)/\lambda. \quad (2)$$

Define the literal carrier test

$$\text{Car}(A) \iff \exists c \in \{1, 2, 3, 6\} : c \mid A, \quad A/c > 3 \text{ prime}.$$

It returns a unique  $(c, q)$  when true. Primality is a finite exact test, not an assumed supply condition. For an integer  $n = \prod_\ell \ell^{e_\ell}$ , also write

$$K(n) = \prod_\ell \ell^{\lceil e_\ell/2 \rceil};$$

this is the least positive integer with  $n \mid K(n)^2$ .

Put  $\delta_E = 4hr\lambda - 1$ . Define  $\mathcal{I}_E(W)$  to be empty unless  $r\lambda \mid 6$  and  $\delta_E$  is squarefree. Otherwise it is exactly

$$\left\{ t \mid \text{oddpair}(s) : \delta_E t^2 < p, \quad \text{Car} \left( \frac{p + \delta_E t^2}{4} \right) \right\}. \quad (3)$$

Also set

$$\mathcal{V}(W) = \{hr^2, hs^2\} \cap \text{Div}(36).$$

For  $\nu \in \mathcal{V}(W)$ , define  $\mathcal{I}_{M,\nu}(W)$  to be empty unless  $R$  is squarefree. The two possible seeds obey

$$\frac{p + 4hr^2}{R} = 4hr\lambda - 1, \quad \frac{p + 4hs^2}{R} = 4hs\lambda - 1.$$

Consequently, for every  $\nu \in \mathcal{V}(W)$  the quantity  $A_\nu = (p + 4\nu)/R$  is a positive odd integer. If  $R$  is squarefree, use

$$\left\{ t \mid A_\nu : Rt^2 < p, \quad a_t = \frac{p + Rt^2}{4} = c_t q_t, \quad c_t \mid 6, \quad q_t > 3 \text{ prime}, \quad \nu \mid c_t^2 \right\}. \quad (4)$$

All such  $t$  are odd. The final budget is retained explicitly even where it also follows from the other integer conditions.

**Theorem 3.1** (Complete bare fibre). *The complete ordered rational mixed-source fibre of  $W$  is the disjoint union of the following records. For each  $t \in \mathcal{I}_E(W)$ , set*

$$R_t = \delta_E t^2, \quad a_t = (p + R_t)/4, \quad (h_t, r_t, s_t) = \left( \frac{a_t}{r\lambda}, \lambda, r \right), \quad u_t = h_t \lambda^2.$$

*Take both original  $E$  tail orders. For each  $\nu \in \mathcal{V}(W)$  and  $t \in \mathcal{I}_{M,\nu}(W)$ , take the two  $M$  words*

$$R_t = Rt^2, \quad a_t = (p + R_t)/4, \quad u_t = \nu \quad \text{or} \quad a_t^2/\nu.$$

*Thus, exactly,*

$$n_W = |\rho^{-1}(W)| = 2|\mathcal{I}_E(W)| + 2 \sum_{\nu \in \mathcal{V}(W)} |\mathcal{I}_{M,\nu}(W)|. \quad (5)$$

*Every original coordinate, prime exponent and ordering of each enumerated preimage is recovered by these formulas and gcd normalization. The maps are defined on arbitrary existing sorted  $M$  targets; an empty fibre is an allowed, completely decided result.*

*Proof.* In the  $E$  return (1), the primitive target parameters force  $(r_t, s_t) = (\lambda, r)$ ,  $H = h$ ,  $J = s$  and  $\delta = \delta_E$ . The identity  $p\lambda + r = \delta_E s$  follows directly from (2). For the reconstructed source one has

$$r(4u_t + 1) = \delta_E(s + \lambda t^2), \quad \gcd(r, \delta_E t^2) = 1.$$

Indeed,  $\delta_E \equiv -1 \pmod{r}$  and  $t \mid s$  with  $(r, s) = 1$ . Multiplication by the unit  $r$  therefore shows that the squared trace gate at  $\delta_E t^2$  is equivalent to

$$\delta_E t^2 \mid (\delta_E s)^2.$$

At a prime of the squarefree  $\delta_E$ , this is  $v_\ell(t) \leq v_\ell(s)$ ; the same inequality holds at every other prime. Oddness gives exactly  $t \mid \text{oddpart}(s)$ . Conversely, that divisibility gives the full squared gate. Since  $4r\lambda \mid 24$ ,  $p \equiv 1$  and  $\delta_E \equiv -1 \pmod{4r\lambda}$ . Moreover  $\lambda R = r + s$  and  $(r, s) = 1$  give  $(s, r\lambda) = 1$ ; hence  $t \mid \text{oddpart}(s)$  gives  $(t, 4r\lambda) = 1$ . Every unit modulo a divisor of 24 has square one. Thus  $t^2$  preserves the source shell congruence. The displayed  $h_t$  is integral,  $u_t \mid a_t^2$ , and (1) returns  $W$ . The carrier test is precisely the extra restriction from the general automatic-ray source to the mixed source.

In the  $M$  return the target residual is the squarefree part of the old residual. Its two possible original divisor orientations are exactly  $hr^2, hs^2$ . The small pre-complement word must be one of these and divide  $c_t^2 \mid 36$ , which gives  $\nu$  and its displayed budget. Put  $N = p + 4\nu = RA_\nu$ . Since  $R$  is squarefree, the gate  $Rt^2 \mid N^2$  is equivalent prime by prime to  $t \mid A_\nu$ . For  $\nu = hr^2$  or  $hs^2$ , respectively,  $K(\nu)$  divides  $hr$  or  $hs$ ; hence  $A_\nu \equiv -1 \pmod{4K(\nu)}$ . Therefore  $t$  is a unit modulo  $4K(\nu) \mid 24$  and  $t^2 \equiv 1 \pmod{4K(\nu)}$ . The explicit condition  $\nu \mid c_t^2$  retains the literal carrier budget.

For the complementary word  $u'_t = a_t^2/u_t$ , first  $\gcd(a_t, R_t) = 1$ : a common divisor would divide  $4a_t - R_t = p$  while  $0 < R_t < p$ . Since  $u_t \mid a_t^2$ , also  $\gcd(u_t, R_t) = 1$ . Thus

$$p + 4u'_t \equiv a_t u_t^{-1}(p + 4u_t) \pmod{R_t},$$

and the multiplier is a unit. The squared  $M$  gates for the complementary words are therefore equivalent. The returned small word is  $\nu$  at  $a$ , so its sorted target is  $W$ .

Every source in either list is a positive rational trace by (1E)–(1M). Each list has the displayed inverse. Distinct  $t$  give distinct old residuals. The two  $M$  words are different because one contains the carrier  $q_t^2$  and the other does not. Different  $\nu$  cannot duplicate an  $M$  source: its unique  $q_t$  determines which of  $u_t, a_t^2/u_t$  is the small word.  $E$  and  $M$  sources are disjoint by the  $p$ -valuation pair of their original factor coordinates. The two  $E$  orders differ because equal tails would require  $s_t = pr_t$ , while  $s_t \leq a_t < p$ . This proves disjointness and completeness.  $\square$

The formula is an executable finite inverse. In (3) use  $t^2 \leq (p-1)/\delta_E$ ; in (4) use  $t^2 \leq (p-1)/R$ . Every prime-factor occurrence in either divisor list is retained. There is no fixed-progression or subgroup-support replacement.

**3.1. The missing section and channel collisions.** The general-ray section  $t = 1$  need not be in the mixed source. At

$$W = (1009; 255, 24216, 686120),$$

the full bare fibre consists of the M words 9, 11449 at  $a_t = 321 = 3 \cdot 107$ ,  $R_t = 275$ ,  $t = 5$ . The squarefree representative  $a = 255$  has no mixed carrier. Thus its fibre has two proper inputs and no  $t = 1$  member. The E-origin example

$$W = (1129; 308, 3387, 1043196)$$

has only the two E orientations at  $a_t = 417 = 3 \cdot 139$ ,  $t = 7$ ; its would-be E-branch section has  $a_t = 285$ , again outside the mixed domain.

An actual hard-prime cross-channel collision is

$$W = (6841; 1866, 20523, 12765306).$$

It has the two E orientations at  $a_t = u_t = 1713 = 3 \cdot 571$  and the two M words 3, 1160652 at  $a_t = 1866 = 6 \cdot 311$ . The four original source records produce the same bare sorted target, not four new targets.

For fixed  $p$ , let  $\mathcal{S}_p$  be the ordered mixed-source records and write  $F_W = \rho^{-1}(W)$  for  $W$  in the actual image. The undecorated free abelian pushforward

$$\Phi_p : \mathbb{Z}[\mathcal{S}_p] \longrightarrow \mathbb{Z}[\text{Im } \rho_p], \quad e_T \longmapsto e_{\rho(T)},$$

has kernel basis  $e_T - e_{T_W}$  for  $T \in F_W \setminus \{T_W\}$ , where  $T_W$  is chosen from that explicitly computed nonempty fibre. Its rank is  $\sum_W (n_W - 1)$ . For the decorated map  $e_{(T, \xi)} \mapsto e_{(W, \xi)}$ , with all  $\xi \in \Xi_W$ , the rank is instead  $16 \sum_W (n_W - 1)$ . Neither section uses an unavailable  $t = 1$  source.

Fix now one frame  $\xi \in \Xi_W$ . With  $O(W, \xi) : V_{\text{labels}} \rightarrow E_{\text{recv}}$ , define

$$L_{W, \xi} : \bigoplus_{T \in F_W} V_{\text{labels}} \longrightarrow E_{\text{recv}}, \quad (v_T)_T \longmapsto O(W, \xi) \sum_T v_T.$$

For positive-definite Hermitian forms  $G_T$ , put  $M_W = \sum_T G_T^{-1}$ . The exact quotient Gram operator is

$$Q_{W, \xi} = O(W, \xi)^{-*} M_W^{-1} O(W, \xi)^{-1} = [O(W, \xi) M_W O(W, \xi)^*]^{-1}. \quad (6)$$

The unique minimum-energy preimage of  $z$  is

$$v_T = G_T^{-1} M_W^{-1} O(W, \xi)^{-1} z.$$

For  $G_T = I$ , this becomes  $v_T = n_W^{-1} O(W, \xi)^{-1} z$ . This complex vector-space section is generally neither an integral coefficient section nor a choice of one arithmetic source. The formulas follow by a Lagrange multiplier or direct completion of the square; they change no original metric by decree.

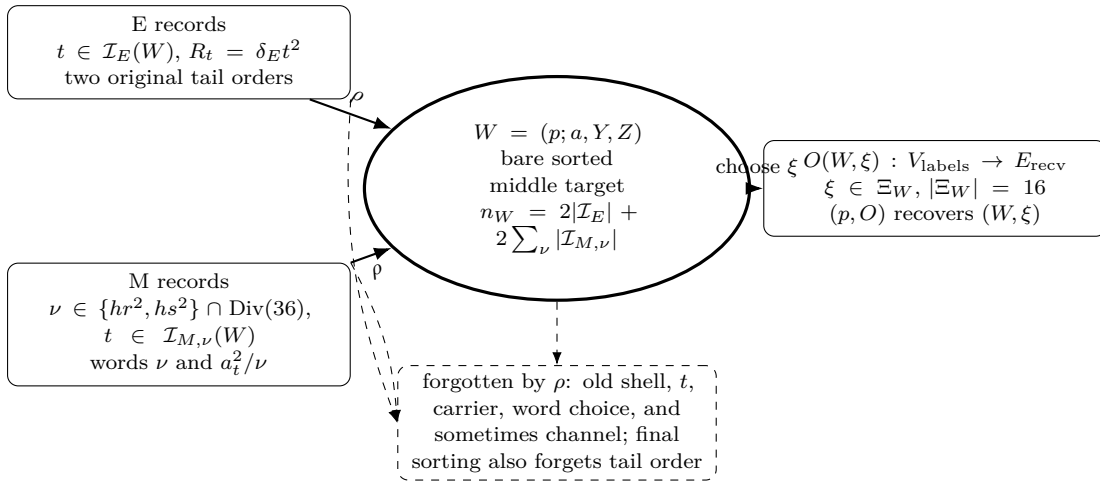


FIGURE 1. The exact arithmetic fibre and receiver lift. The E and M records are disjoint as source records but may meet at one target. Adding the full matrix restores  $W$  and the selected square-root frame, not the forgotten arithmetic record. The fibre theorem and receiver reconstruction formulas give the exact maps.

## 4. A RECEIVER MATRIX ITSELF RETAINS THE TARGET

Retain the literal roots  $\ell = (p, a, Y, Z)$ ,  $S = \sum \ell_j$  and

$$H(X) = -S^{-1} \prod_j (X - \ell_j), \quad A = -S^{-1}, \quad d_j = H'(\ell_j), \quad \xi_j^2 d_j = 1.$$

The original receiver  $O : V_{\text{labels}} \rightarrow E_{\text{recv}}$ , with its sixteen full-frame square-root choices, has columns

$$O_{\cdot j} = \xi_j (1, -id_j, Ad_j^2 + 2\ell_j d_j, i(7\ell_j^2 d_j - 13d_j^2))^{\top}. \quad (7)$$

Its invertibility on actual integral witnesses is the current global-sign and sharp-growth theorem, not an additional assumption (The Clankers, 2026c).

**Proposition 4.1** (Fixed-prime target reconstruction). *The map  $(W, \xi) \mapsto (p, O)$  is injective on the marked target domain. Its explicit inverse is*

$$\begin{aligned} \xi_j &= O_{0j}, & d_j &= iO_{1j}/O_{0j}, \\ A &= \frac{O_{20}/O_{00} - 2pd_0}{d_0^2}, & \ell_j &= \frac{O_{2j}/O_{0j} - Ad_j^2}{2d_j}. \end{aligned} \quad (8)$$

The fourth row, sum normalization, square-root equations,  $d_j = A \prod_{k \neq j} (\ell_j - \ell_k)$  and the arithmetic domain give the remaining exact consistency tests.

*Proof.* All displayed denominators are nonzero. Dividing each receiver column by its retained first coordinate gives  $d_j$  and then the affine expression in  $A, \ell_j$ . The known prime is the root with label zero, determining  $A$ . The remaining equations solve for each original root. Substitution into (7) proves both compositions. No unlabelled norm or singular spectrum has been used instead of the matrix.  $\square$

Thus Theorem 3.1 also gives the complete arithmetic fibre of the bare fixed-prime receiver-valued composite for each selected square-root frame. The matrix recovers the returned sorted target and those four branches, not a unique choice among its  $n_W$  arithmetic preimages. Root permutation and branch quotienting are not additional unrecorded identifications. This target reconstruction does not create a missing target.

## 5. A FIXED-RESIDUAL MIDDLE FAMILY AND EVERY FIRST CORRECTION

Fix positive integers  $R \equiv 3 \pmod{4}$  and  $u > 0$ , with  $\gcd(R, u) = 1$ . At the prime values satisfying the original conditions

$$a = (p + R)/4 \in \mathbb{Z}, \quad u \mid a^2, \quad R \mid p + 4u,$$

retain the middle witness, sorted for sufficiently large  $p$ ,

$$(a, Y, Z) = \left( a, \frac{p(a+u)}{R}, \frac{pa(a+u)}{Ru} \right). \quad (9)$$

The identity holds at every positive real parameter before these arithmetic tests; the asymptotic derivation below uses this exact rational family. For  $p > 4(R+u)$  the literal order is  $a < p < Y < Z$ , so the derivative signs and branch convention below already hold on an explicit half-line. Define

$$\alpha = \frac{1}{4R}, \quad \beta = \frac{1}{16Ru}, \quad C = \sqrt{\frac{3}{16R}}.$$

Take positive real  $\xi_a, \xi_Y$ , and  $\xi_p = i/\sqrt{-d_p}$ ,  $\xi_Z = i/\sqrt{-d_Z}$ , in original label order. Then  $\epsilon = A^2 \prod_{i < j} (\ell_j - \ell_i) \prod_j \xi_j = 1$ .

**Theorem 5.1** (Middle spectrum with first corrections). *For fixed  $R, u$ , as  $p \rightarrow \infty$  along this exact family,*

$$\begin{aligned} s_1 &= 20\beta^3 p^9 \left[ 1 + \frac{6R + 14u/5}{p} + O_{R,u}(p^{-2}) \right], \\ s_2 &= 2\alpha^2 p^4 \left[ 1 + \frac{-R/2 + 19u/5}{p} + O_{R,u}(p^{-2}) \right], \\ s_3 &= \sqrt{2}C p^{3/2} \left[ 1 - \frac{41R}{12p} + O_{R,u}(p^{-2}) \right], \\ s_4 &= \frac{\sqrt{2}}{C} p^{-3/2} \left[ 1 + \frac{11R}{12p} + O_{R,u}(p^{-2}) \right], \end{aligned}$$

and

$$\det O = 80\alpha^2 \beta^3 p^{13} \left[ 1 + \frac{3R + 33u/5}{p} + O_{R,u}(p^{-2}) \right]. \quad (10)$$

In particular,

$$p^{-3/2} O^{-1} \longrightarrow \frac{C}{2} (-i, 1, 0, 0)^\top e_0^*,$$

$$p^{-3/2} \|O^{-1}\|_2 \rightarrow \sqrt{3/(32R)}, \quad \|\wedge^2 O^{-1}\|_2 \rightarrow \frac{1}{2}, \quad p^4 \|\wedge^3 O^{-1}\|_2 \rightarrow 4R^2.$$

Every remainder is for fixed  $R, u$ . No claim uniform in unbounded  $R, u$  is made, and a value of  $p$  used in an analytic check is not thereby declared prime.

**5.1. The complete analytic matrix.** Let  $z = 1/p$  and define

$$B(z) = \text{diag}(1, p^{-3}, p^{-4}, p^{-6}) O_p \text{diag}(p^{3/2}, p^{3/2}, 1, p^{-3}).$$

This is an explicitly compensated calculation, not an identification of the original norms under rescaling. Its inverse scaling is retained below. Directly from the rational roots and  $H'$ , the derivative expansions are

$$\begin{aligned} d_p &= -C^2 p^3 [1 - 10R/(3p) + O(p^{-2})], & d_a &= C^2 p^3 [1 - R/(3p) + O(p^{-2})], \\ d_Y &= \alpha^2 p^4 [1 - 3R/p + O(p^{-2})], & d_Z &= -\beta^2 p^6 [1 + 4R/p + O(p^{-2})]. \end{aligned}$$

For example, the first coefficients of the three differences at the prime root are  $-R/3, -3R + 4u, 2R + 4u$ , while the relative first coefficient of  $S$  is  $2R + 8u$ . Their sum minus the latter is  $-10R/3$ . The same difference calculation gives the other three coefficients.

After factoring these derivative leading terms out of the square roots, every remaining radical is analytic and equal to one at  $z = 0$ . Every entry of  $B$  then has a nonnegative integral valuation. Thus  $B(z) = B_0 + zB_1 + O_{R,u}(z^2)$ , where

$$\begin{aligned} B_0 &= \begin{pmatrix} i/C & 1/C & 0 & 0 \\ -C & -iC & 0 & 0 \\ -2iC & C/2 & 2\alpha^2 & 0 \\ 13C^3 & -13iC^3 & -6i\alpha^3 & 20\beta^3 \end{pmatrix}, \\ B_1 &= \begin{pmatrix} 5Ri/(3C) & R/(6C) & 0 & 0 \\ 5RC/3 & iRC/6 & -i\alpha & 0 \\ i(10RC/3 - C^3/\beta) & 5RC/12 - C^3/\beta & (-R + 4u)\alpha^2 & -3i\beta^2 \\ 7C - 65RC^3 & i(13RC^3/2 + 7C/16) & i(62R + 56u)\alpha^3 & (120R + 56u)\beta^3 \end{pmatrix}. \end{aligned}$$

These coefficients follow from  $(1 + ez)^{-1/2} = 1 - ez/2 + O(z^2)$  and the four receiver polynomials in (7). Every original column, including its zero leading entries and next coefficient, is shown. An explicit analytic neighbourhood can be computed from the finitely many normalized numerator and denominator polynomials of the derivative units: for each  $P(0) = 1$  choose  $\rho \leq [4(1 + \sum_{k>0} |[z^k]P|)]^{-1}$ . Then  $|P(z) - 1| < 1/4$  for  $|z| \leq \rho$ . The chosen square-root branches are analytic there; continuity and  $\det B_0 \neq 0$  give an inverse neighbourhood. The printed  $O_{R,u}$  terms do not assert a uniform numerical neighbourhood.

**5.2. Complete compound calculation and error control.** One has  $\det B_0 = 80\alpha^2\beta^3$ . At exterior ranks 1, 2, 3, 4, the leading norms and their relative first coefficients are

| $k$ | power of $p$ | leading norm                 | first coefficient |
|-----|--------------|------------------------------|-------------------|
| 1   | 9            | $20\beta^3$                  | $6R + 14u/5$      |
| 2   | 13           | $40\alpha^2\beta^3$          | $11R/2 + 33u/5$   |
| 3   | 29/2         | $40\sqrt{2}C\alpha^2\beta^3$ | $25R/12 + 33u/5$  |
| 4   | 13           | $80\alpha^2\beta^3$          | $3R + 33u/5$      |

Here the first rank uses row 3, column  $Z$ . The second uses rows 2,3 and columns  $Y, Z$ . The third uses the complete row 1,2,3 on both triples  $(p, Y, Z)$  and  $(a, Y, Z)$ ; their leading squared norms are equal and their relative first coefficients are  $-R/6 + 33u/5$  and  $13R/3 + 33u/5$ . Their average gives the displayed correction. The fourth uses  $\text{tr}(B_0^{-1}B_1) = 3R + 33u/5$ .

To check that no other minor changes those coefficients, the row powers of  $O$  are  $(0, 3, 4, 6)$  and the column powers are  $(-3/2, -3/2, 0, 3)$ . Add the appropriate powers for each minor, reducing its order by at least one when its constant  $B$ -minor vanishes. At rank 1 all other entries are  $O(p^6)$ ; at rank 2 all other entries are  $O(p^{12})$ ; at rank 3 all other entries are  $O(p^{13})$ . The finite constant/linear minor table supplied in the certificate includes every row and column combination. The leading entry or leading row gives a lower bound for the operator norm; the full Frobenius norm gives the matching upper bound. The squares of the omitted entries contribute relative  $O(p^{-2})$  or smaller. Thus the displayed first coefficients concern the entire exterior operators, not selected minors. Successive ratios of compound norms prove Theorem 5.1.

For a second direct calculation, put  $z_0 = B_0^{-1}e_0$  and  $z_1 = -B_0^{-1}B_1z_0$ . They are exactly

$$z_0 = (-iC/2, C/2, 9R/8, 576iRu^3/5)^\top,$$

$$z_1 = \left( -2iRC/3, -19RC/12, \frac{3R(25R - 228u)}{80}, \frac{24iRu^3(-1160R - 717u)}{25} \right)^\top.$$

Their defining equations verify all coordinates. The original inverse is

$$O^{-1} = \text{diag}(p^{3/2}, p^{3/2}, 1, p^{-3})B(1/p)^{-1}\text{diag}(1, p^{-3}, p^{-4}, p^{-6}). \quad (11)$$

In particular, its four first-column powers are  $p^{3/2}, p^{3/2}, 1, p^{-3}$ , not the exterior fixed- $c$  family's  $p^{3/2}, p^{3/2}, 1, p^{-6}$ .

## 6. THE MIDDLE OBSERVATION HIERARCHY IS 0, 2, 3, 9

Let  $v_p$  be the right singular vector for  $s_4(O)$ , with its  $a$  coordinate positive. Set

$$v_0 = (-i, 1, 0, 0)^\top/\sqrt{2}, \quad w_0 = (i, 1, 0, 0)^\top/\sqrt{2}, \quad n_0 = C/\sqrt{2}.$$

Then

$$v_p = v_0 - \frac{9R}{4p}w_0 + \frac{9R}{8n_0}p^{-3/2}e_Y + O_{R,u}(p^{-2}),$$

and the two tail components have the stronger relative expansions

$$e_Y^*v_p = \frac{9R}{8n_0}p^{-3/2} \left[ 1 + \frac{7R/4 - 38u/5}{p} + O_{R,u}(p^{-2}) \right], \quad (12)$$

$$e_Z^*v_p = \frac{576iRu^3}{5n_0}p^{-9/2} \left[ 1 + \frac{-35R/4 - 239u/40}{p} + O_{R,u}(p^{-2}) \right].$$

To prove the last, small component rather than infer it from a larger vector error, divide (11) by  $p^{3/2}$  and write

$$D(z)B(z)^{-1}E(z), \quad D = \text{diag}(1, 1, z^{3/2}, z^{9/2}), \quad E = \text{diag}(1, z^3, z^4, z^6).$$

The right Gram has a simple leading eigenvalue  $C^2/2$ . Its complementary block equations give its top eigenvector proportional to  $(1, O(z^3), O(z^4), O(z^6))$ . Multiplication by  $E$  leaves the coefficient  $e_0 + O(z^6)$ . After return to label coordinates the vector is therefore  $D(z)[B(z)^{-1}e_0 + O(z^6)]$ . Its norm is  $n_0[1 - 11Rz/12 + O(z^2)]$ . Substitution of  $z_0, z_1$  proves all displayed coefficients, including the relative error in the last coordinate.

Also

$$p^3 s_4^2 = \frac{32R}{3} + \frac{176R^2}{9p} + O_{R,u}(p^{-2}).$$

For every fixed nonzero scalar row  $L$  and  $\tau > 0$ , put  $\mathcal{H}_p(\tau) = \exp(-\tau p^3 O^* O)$ . The complete hierarchy is

| successive original-label tests | $\nu$ | $\lim p^\nu L \mathcal{H}_p(\tau) L^* / e^{-(32R/3)\tau}$ |
|---------------------------------|-------|-----------------------------------------------------------|
| $Lv_0 \neq 0$                   | 0     | $ Lv_0 ^2$                                                |
| $Lv_0 = 0, Lw_0 \neq 0$         | 2     | $(81R^2/16) Lw_0 ^2$                                      |
| $Lv_0 = Lw_0 = 0, Le_Y \neq 0$  | 3     | $(27R^3/2) Le_Y ^2$                                       |
| $Lv_0 = Lw_0 = Le_Y = 0$        | 9     | $(3538944R^3u^6/25) Le_Z ^2$                              |

The four directions are a basis, so the tests exhaust every nonzero row. The other three heat eigenvalues are  $O_{R,u}(e^{-c\tau p^6})$  for a positive constant at fixed  $R, u$ ; their contribution stays negligible after these polynomial scalings. Convergence is locally uniform for  $\tau \in (0, \infty)$ , not at zero time. Fixed sign changes act in the actual label frame and transform the tests; phases cannot be replaced by moduli.

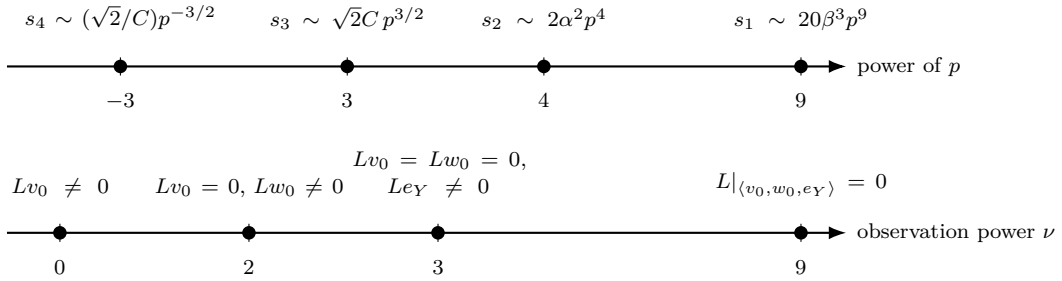


FIGURE 2. Two different exact exponent systems for the fixed- $(R, u)$  middle family. The upper axis records the four singular-value scales. The lower axis records the exhaustive rescalings of  $L \exp(-\tau p^3 O^* O) L^*$  for fixed  $\tau > 0$ ; its last direction is  $e_Z$ . The first corrections and coefficients are stated in the middle-spectrum theorem and the subsequent observation table.

For fixed positive original metrics  $G, Q$ , the orthonormal representative  $\hat{O} = Q^{1/2} O G^{-1/2}$  has inverse  $\hat{O}^{-1} = G^{1/2} O^{-1} Q^{-1/2}$ . Its leading rank-one limit follows from Theorem 5.1 with both factors retained. Writing  $a_0 = C(-i, 1, 0, 0)^T/2$ , the limiting small energy is  $[(a_0^* G a_0)(e_0^* Q^{-1} e_0)]^{-1}$ . A conductor replaces  $Q$  by its actual pullback; neither unitary transport nor uniform moving-metric constants are inferred. The detailed fixed-metric method is inherited from ES33–ES37 in The Clankers (2026d).

## 7. ALL THREE UPPER EXPONENTS ARE SHARP ON HARD MIDDLE WITNESSES

The current uniform theorem gives, with  $\gamma = 80/6279^4$ , the M bounds

$$\|O^{-1}\|_2 < \frac{224}{\gamma} p \sqrt{w}, \quad \|\wedge^2 O^{-1}\|_2 < \frac{180}{\gamma}, \quad \|\wedge^3 O^{-1}\|_2 < \frac{41}{4\gamma p^2}, \quad w = p/(4a - p) \leq p/3$$

(The Clankers, 2026c, Theorem 1.1 and middle cofactor proof). We now restrict optimality to actual middle states at  $p \equiv 1 \pmod{840}$ .

**7.1. First hard progression.** Choose  $R = 11, u = 9$  and primes

$$p \equiv 8401 \pmod{9240}.$$

They satisfy  $p \equiv 1 \pmod{840}$ ,  $p + 36 \equiv 0 \pmod{11}$  and  $3 \mid a = (p + 11)/4$ , so every member has the original M state (9). The residue is coprime to 9240, hence Dirichlet's theorem supplies infinitely many such primes (Dirichlet, 1837, conclusion at source lines 1257–1265); see also Sutherland (2017, Theorem 18.1, p. 1). Theorem 5.1 proves positive limits at inverse scales  $p^{3/2}$  and 1 for its first and second compounds. Its initial prime sample  $p = 26881$  has  $a = 6723 = 3^4 \cdot 83$ ,

$R = 11$ ,  $u = 9$ . It has no mixed input fibre:  $r\lambda = 68$  excludes the E list, and  $(p + 36)/11 = 2447$  is prime, so only  $t = 1$  fits the M cutoff, with an unavailable mixed carrier. This is a concrete reason not to assign these optimality claims to the narrower image.

**7.2. Second hard progression.** For primes

$$p \equiv 4201 \pmod{9240}, \quad s = (p + 1)/11,$$

use the original M state

$$(h, r, s, \lambda) = (3, 1, (p + 1)/11, 1), \quad (a, Y, Z) = (3s, 3p, 3ps).$$

The residue is again coprime to the modulus and is 1 (mod 840). The identity is immediate from  $p = 11s - 1 = 12s - s - 1$ , and the denominators are sorted for these primes. Put  $\gamma_0 = 3/11$ . The three small root ratios are 1,  $3/11$ , 3 and the large root is  $Z = \gamma_0 p^2 [1 + O(p^{-1})]$ . The derivative leading coefficients at those small roots are

$$D_0 = -16/11, \quad D_1 = 240/121, \quad D_2 = 60/11.$$

Take the same positive-real or positive-imaginary branch convention according to their signs. In rows 0, 1, 2, the three core columns, after the respective row powers  $p^{-1}, p, p^2$ , have the matrix

$$C_* = \begin{pmatrix} i\sqrt{11}/4 & 11\sqrt{15}/60 & \sqrt{165}/30 \\ -4\sqrt{11}/11 & -4i\sqrt{15}/11 & -2i\sqrt{165}/11 \\ -8i\sqrt{11}/11 & 24\sqrt{15}/121 & 12\sqrt{165}/11 \end{pmatrix}.$$

Its last row has squared norm  $2968784/14641$ , the exterior norm squared of its last two rows is  $73267200/161051$ , and  $\det C_* = 2896/121$ . These constants follow by the displayed three-by-three products; their Gaussian-rational, square-root-cancelled versions are in the certificate.

The first prime in this progression is  $p = 4201$ , giving  $(a, Y, Z) = (1146, 12603, 4814346)$ . Its mixed-source fibre has exactly the two M words 3 and 437772; this finite occupancy is not used to infer occupancy for the progression.

The original  $Z$  column has last entry  $20\gamma_0^3 p^6 [1 + O(p^{-1})]$ . Its other entries are of orders  $p^{-2}, p^2, p^4$ , while the three core columns have orders  $p^{-1}, p, p^2, p^3$ . Thus the complete leading compound constants are

$$20\gamma_0^3 p^6, \quad 20\gamma_0^3 \sqrt{2968784/14641} p^8, \quad 20\gamma_0^3 \sqrt{73267200/161051} p^9, \quad 20\gamma_0^3 (2896/121) p^8.$$

For ranks two and three, all the entries involving the leading  $Z$  last coordinate form one row, with respectively all core row-2 entries and all core rows-1,2 minors. Every other term is of lower power. For rank four, eliminating the  $Z$  column changes the core rows only in lower powers. The same operator/Frobenius comparison therefore proves

$$p^{-1} \|O^{-1}\|_2 \rightarrow \sqrt{286200/360371}, \quad \|\wedge^2 O^{-1}\|_2 \rightarrow \frac{\sqrt{2968784}}{2896}, \quad p^2 \|\wedge^3 O^{-1}\|_2 \rightarrow \frac{121}{2896}. \quad (13)$$

All constants are nonzero. Along the two reduced prime progressions, these limits prove that no smaller power than  $p^{3/2}$ , no decaying bound replacing  $O(1)$ , and no faster power decay than  $p^{-2}$  is possible in the inherited three uniform estimates, already on hard middle witnesses. This does not assert prime-carrier preimages for these families; their membership in the narrower image of  $\rho$  is determined separately by Theorem 3.1. No unproved simultaneous primality condition is used.

## 8. VERIFICATION, SOURCE BOUNDARIES, AND THE REMAINING THEOREM

The main arithmetic replay enumerates all original first-half divisor words at primes  $p \equiv 1 \pmod{24}$  through 10000. It computes all sorted M targets and compares their formula (5) fibres, including empty fibres, against the full forward mixed source. The separate checker imports neither it nor any predecessor code; it uses positive factor pairs of  $(pa)^2$  for rational inputs and primitive  $h, r, s$  enumeration for integral targets. The exact receiver jets are independently expanded from their literal rational formulas in standard-library Gaussian-rational Laurent arithmetic; all leading-compound positions are retained. For the second hard family, the checker verifies the entered leading  $x, D$  data and resulting constants, while the written calculation derives those

data, the sign and the dominance from the full  $p$ -dependent receiver. The parameter-grid checks corroborate the written analytic derivation, not an extrapolated prime theorem.

```
python run_all.py --directory reproduced
python fibres.py --target 1009 255 24216 686120
```

The executed counts, commands and hashes are recorded in the receipts. There is no independent human mathematical review or Lean build.

The completed source-fibre calculation repairs no false universal claim: it fills exactly the explicit M-origin gap left by the corrected composite. The hard-prime middle asymptotics refine a post-witness result; they do not reverse receiver invertibility into original source occupancy. The still-unproved statement remains that every prescribed hard prime has an original E or M state. Neither a nonempty mixed fibre at every M target, nor a mixed source at every prime, is asserted. The programme remains at Turn 7 on that existence obligation.

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